Speech-To-Text and NLP: Comparing De Gasperi and Meloni's Rhetoric

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This report compares the rhetoric of Alcide De Gasperi and Giorgia Meloni. After building a corpus of public speeches transcribed with Whisper from Giorgia Meloni's videos on YouTube, I compare them with a corpus of speeches from the Italian Republic's first Prime Minister Alcide De Gasperi. The results show that De Gasperi's speeches are, on average, 30% more diversified in their lexicon, and the corpora have a Jaccard similarity index of 32%. The highest-frequency words in common show transversal themes, such as freedom, peace and nation, while the highest-frequency words not in common reflect the PMs' different political and societal landscapes. I also performed a sentiment analysis on each speech using a BERT classifier. The results underscore the importance of domain-appropriate training data: the model was trained on product reviews, not on political rhetoric, which translated to an average confidence of roughly 40% per classification.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Speech recognition**; • **Social and professional topics** → **Political speech**; • **Information systems** → **Sentiment analysis**.

Additional Key Words and Phrases: AI, NLP, Politics, Speeches

1 INTRODUCTION

In traditional photography, image acquisition is a direct process: a physical lens focuses light reflected from an object onto a photosensitive sensor, comprised by billions of photosites each of which records the raw intensity values of light in three channels: Red, Green, and Blue. The post-processing needed to then obtain a polished-looking picture are quite straightforward, since the object of interest is the *external* appearance of the subject.

On the other hand, many modern scientific and medical apparatuses require looking inside an object to capture phenomena where direct visual access is not possibile, for example when taking an X-Ray of a patient. In this paradigm, the raw hardware measurements are often highly degraded, indirect, or completely unrecognizable, and the final image must be computationally reconstructed. This constraint gave rise to the interdisciplinary field of *Computational Imaging* (CI), which leverages machine learning and other paradigms to overcome hardware limitations and reconstruct high-quality images from raw, unstructured sensor data. This represents a paradigm shift in visual and diagnostic modalities, transitioning to an integrated approach that couples physical measurement devices with rigorous data-driven and variational algorithms.

An example of an application of computational imaging is MRI subsampling: the idea is to purposely acquire only a fraction of the raw data to speed-up the process of a factor of 2X to 10X, to minimize claustrophobia in patients. The image is then reconstructed from the incomplete data.

Mathematically, the core of Computational Imaging lies in the distinction between the *forward problem* and the *inverse problem*. The forward problem models the underlying physics of the imaging system. Assuming the true, continuous underlying object is represented by u, and the discrete, noisy measurements recorded by the hardware are y, the acquisition process can be formulated as:

$$y = Au + \eta \tag{1}$$

where A represents the forward operator (e.g., a blurring kernel, a Fourier transform, or a Radon transform) dictated by the system's physics, and η denotes additive measurement noise. The objective of Computational Imaging is to solve

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the corresponding *inverse problem*: given the corrupted measurements y and the known or estimated forward operator A, find an approximation u^* that accurately recovers the true state of the object.

A classic and highly illustrative archetype of such an inverse problem is Computed Tomography (CT). A CT scanner does not capture a direct image of internal anatomy; instead, it measures the attenuation of X-ray beams passing through the body at various angles, yielding raw data known as a sinogram. Reconstructing a cross-sectional image from this sinogram is inherently a computational task. While classical methods like Filtered Backprojection (FBP) succeed when a dense, continuous set of projection angles is available, real-world constraints—such as the urgent clinical need to minimize X-ray radiation doses or accelerate scanning times—frequently limit data acquisition. This results in *Limited Data Tomography*, where the inverse problem becomes highly ill-posed, meaning the physical measurements alone are mathematically insufficient to yield a unique, stable reconstruction without introducing severe structural artifacts.

This essay explores the paradigm shifts within Computational Imaging through the practical application of limited-data CT image reconstruction, reflecting the algorithmic journey from classical mathematical optimization to modern, data-driven methodologies. First, we examine *model-based* variational approaches, focusing on the incorporation of hand-crafted image priors—such as Total Variation (TV) regularization—to mathematically compensate for missing angular data. Second, we transition to contemporary *data-driven* frameworks, investigating how deep learning can be integrated with the known physics of the forward operator via physics-informed neural networks or unrolled optimization architectures. By analyzing the empirical results of these two approaches on CT reconstruction tasks, this work highlights how modern Computational Imaging effectively bridges physical modeling and machine learning to overcome the fundamental limits of physical hardware.

2 COMPUTED TOMOGRAPHY

Computed Tomography (CT) is a non-invasive imaging technique that reconstructs the internal structure of an object from measurements taken externally. The term *tomography* itself derives from the Greek words *tomos* (slice or section) and *graphein* (to write or record), which reflect its purpose of producing representations of a target by observing

cross-sectional images of a target without physical intervention [**?**].

The physical principle underlying CT relies on the transmission of waves or particles, most commonly X-rays, through the object of interest. As X-rays traverse matter, their intensity is attenuated according to the material's local properties. In the simplest homogeneous case, this relationship is captured by the exponential decay $I_1 = I_0 e^{-\mu s}$, where I_0 and I_1 denote the incident and transmitted intensities, $\mu > 0$ is the attenuation coefficient, and s is the path length through the material. For non-homogeneous objects, this generalises to the *Beer–Lambert law*:

$$I_1 = I_0 \, e^{-\int_\ell \mu(x)\, dx}, \tag{2}$$

which, upon a logarithmic transformation, yields the line integral relation $-\log(I_1/I_0) = \int_\ell \mu(x)\, dx$. This linearised measurement, referred to as the *absorption* or *projection*, forms the basis of the tomographic inverse problem. A collection of such projections acquired over multiple angles constitutes a *sinogram*.

Mathematically, CT reconstruction can be cast as an inverse problem. Given a target $u \in X = L^2(\Omega)$ with $u\colon \mathbb{R}^d \to \mathbb{R}_+$ defined on a bounded domain $\Omega \subset \mathbb{R}^d$ $(d = 2, 3)$, the *forward problem* models the acquisition process via the Radon transform $\mathcal{R}(u) = y$, which integrates u over lines $\ell(\omega, s)$ parameterised by angle $\omega \in S^1$ and offset $s \in \mathbb{R}$. The reconstruction task, i.e., recovering u from measured data y, constitutes the corresponding *inverse problem*.

The classical analytic solution to this inverse problem is *Filtered Backprojection* (FBP), an exact inversion formula valid in the continuous setting. FBP performs well when complete projection data are available and the target can be assumed static. However, it suffers from theoretical limitations under discretisation and provides no stability guarantees in realistic acquisition conditions.

In many practical scenarios, reducing the X-ray radiation dose or shortening scan time leads to *limited data tomography*, which encompasses two main regimes: *limited-angle CT*, where projections are acquired only within a restricted angular range, and *sparse-angle CT*, where a reduced number of projection angles are sampled over the full range. Both settings cause FBP to produce severely degraded reconstructions, motivating the development of more advanced reconstruction methods capable of handling incomplete and noisy measurement data.

REFERENCES

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